NASA SPACE APPS CHALLENGE 2026 • GLOBAL COMPETITION

AETHER

Planetary & Earth System Trend Detective

Synthesizing 40 Years of NASA Earth Observation Telemetry into an Interactive 3D Digital Twin

FORENSIC ENGINE

- Real-Time WebGL/Three.js at 60 FPS
- 1980–2026 Multi-Decadal Scrubber
- Mann-Kendall Statistical Sieve ($p < 0.001$)
- 3D Planetary Internal Cutaways

OFFICIAL SUBMISSION

- Challenge: Be An Earth System Trend Detective!
- Live Web App: https://aether-nasa.vercel.app
- GitHub: github.com/GabrielCodexxx/AetherNasa
- Team: Aether Detective Team (Gabriel Portillo)

Publicly Accessible • Zero Registration Required • 100% Open Source

THE PROBLEM: DROWNING IN ENVIRONMENTAL NOISE

Why Detecting Genuine Planetary Trajectories is Difficult for Citizens and Scientists

1. Data Fragmentation & Silos
Terabytes of vital telemetry scattered across disparate databases (PODAAC, GES DISC, Earthdata, NSIDC), making holistic system analysis nearly impossible for non-specialists.

2. Signal vs. Noise Dilemma
Short-term weather oscillations (ENSO, seasonal anomalies, volcanic aerosols) mask authentic, decadal climate trends, leading to widespread misinterpretation.

3. The Regional Divergence Paradox
A single underlying forcing (global warming) causes opposite physical responses simultaneously (e.g. Arctic greening vs. Amazon drying; polar ice loss vs. post-glacial rebound).

4. The Public Accessibility Barrier
Complex mathematical formats and gated academic tools prevent students, educators, and journalists from verifying empirical climate evidence directly.

NASA Space Apps 2026 Objective: Answer What, Where, How Much, and Is It Significant?

THE SOLUTION: AETHER 3D DIGITAL TWIN

Instant, Browser-Based 60 FPS Earth System Intelligence at aether-nasa.vercel.app

ORBITAL MECHANICS

- High-fidelity Three.js WebGL at 60 FPS
- Complete solar system context
- Real-time orbital paths for Terra, Aqua, GRACE-FO, ICESat-2, Sentinel-6
- NASA Blue Marble 2K textures

40-YEAR SCRUBBER

- Travel from 1980 to 2026
- Procedural Earth response reflects decadal warming
- Visual polar ice cap retreat (-12.6% per decade)
- Albedo feedback and high-latitude vegetation shift simulation

GEOMETRIC HUD

- Precision glassmorphic HUD
- Zero-circular-dot design rule: diamond telemetry pins
- Real-time anomaly charts and decadal rate indicators
- Procedural Web Audio API soundscape synthesis

Zero Install • Zero Sign-Up • Runs on Mobile, Tablet, Laptop, and 4K Displays

SCIENTIFIC RIGOR & NASA DATA INTEGRATION

Primary Satellite Telemetry Validated Through Non-Parametric Statistics

PRIMARY NASA DATA STREAMS

- NASA GISTEMP v4: Surface temperature anomalies (+1.39°C global mean warming)
- NASA/GFZ GRACE & GRACE-FO: Gravimetric mass balance (-270 Gt/yr Greenland, -135 Gt/yr Antarctic ice loss)
- Sentinel-6 & Jason-3: Radar altimetry (+4.2 mm/yr sea level rise)
- NASA OCO-2 & NOAA GML: Atmospheric CO2 column densities (426 ppm)
- NASA MODIS Terra/Aqua: NDVI vegetation greening/browning

MATHEMATICAL FORENSIC SIEVE

- Mann-Kendall Monotonic Test: Non-parametric rank correlation proving non-random trajectory ($p < 0.001$)
- Sen's Slope Estimator: Robust median-based slope calculation resistant to sensor outliers
- Statistical Significance Locks: Verified at $p < 0.05$, $p < 0.01$, and $p < 0.001$ levels
- Noise Filtering: Prevents temporary seasonal anomalies from triggering false trend alarms

100% Scientifically Vetted: Turning Raw Satellite Data into Defensible Climate Intelligence

BREAKTHROUGH INNOVATION: CUTAWAYS & DIVERGENCE

From Planetary Core Geodynamics to Polar Thermal Amplification

3D PLANETARY INTERIOR CUTAWAYS

- 3/4 cutaway cross-section revealing Earth's Crust, Mantle, Outer Core, and Inner Core
- Interactive geometric callout badges with depth kilometers
- Geodynamic Connection: Illustrates how outer core convection drives the planetary geodynamo
- Connects mantle plumes and tectonic plate boundaries with surface atmospheric retention

DUAL-POLE DIVERGENCE THERMAL LENS

- Visualizes Arctic Amplification: polar warming accelerating at 4x the global planetary average
- Regional Divergence Mapping: contrasting Northern tundra greening against low-latitude forest desiccation
- Altimetry Discrepancies: explains sea level rise in Pacific atolls vs. relative sea drop in post-glacial rebounding Scandinavia

Comparative Planetology: Exploring Why Earth Remains Habitable While Venus and Mars Diverged

CITIZEN SCIENCE & SUPABASE CLOUD SYNC

Empowering Distributed Global Climate Investigation and Classroom Collaboration

DETECTIVE LOGBOOK

- Guided investigation workflow
- Log callsign, variable, target year, and empirical verdict
- Record observational notes directly against satellite data
- Gamified case tracking inspires STEM classroom curiosity

SUPABASE POSTGRESQL

- Live PostgreSQL database on Supabase Edge infrastructure
- Row-Level Security (RLS) protects data integrity
- Global real-time sync across connected research stations
- Instant schema migration with zero manual server maintenance

OFFLINE-FIRST ENGINE

- Graceful degradation to local browser storage
- 100% resilient if internet drops during field expeditions
- Procedural Web Audio anomaly synthesis for auditory learners
- Frictionless export to research memos

Live Cloud Endpoint: toakvrrofklcjwyjahcp.supabase.co • Fully Secured & Synchronized

GLOBAL IMPACT & OPEN SCIENCE SUBMISSION

NASA Space Apps 2026 Submission Package • Deployed and Publicly Accessible

SUBMISSION VERIFICATION

- Challenge: Be An Earth System Trend Detective!
- Live Web App: https://aether-nasa.vercel.app
  (No password, permission, or login required)
- Open-Source Repo: github.com/GabrielCodexxx/AetherNasa
  (MIT License, complete source code & schemas)
- NASA Data Citation: GISTEMP, GRACE-FO, Sentinel-6, OCO-2, MODIS
- Limits Adherence: Exactly 7 Slides / <= 30s Video Walkthrough

INTENDED IMPACT & ROADMAP

- Education: Turn every STEM student into an Earth Detective
- Journalism: Provide fact-checked NASA telemetry briefing cards
- Policy: Bridge the gap between complex science and decision-making
- Roadmap Q1 2027: Integrate NASA SWOT surface water telemetry and live satellite pass alerts

'Uncovering the Trends Shaping Our Living Planet.'

Explore Aether Now at https://aether-nasa.vercel.app • Thank You, NASA Judges!